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Chemical Bonding and Hybridisation — Issue #32 specimen: Construction

Contents and concept route

Concept Book

Chemical Bonding and Hybridisation

Concept route

1. Steric electron-domain counting and multiple bonds
 1. Separate a bonded-atom direction from the number of shared pairs.
 2. The counting centre is nitrogen, not every atom in NF_3 .
 3. Critique the counting operation, rather than merely giving a corrected label.
2. Hybridisation, orbital conservation, and overlap symmetry
 1. Distinguish the hybrid-set name from the number of orbitals in that set.
 2. The task asks for a local count before the introductory geometry label.
 3. Use the stated internuclear axis to classify the overlap.
 4. Map each set name to an orbital inventory, not its superscript alone.

3. 3D Stereochemical representation and qualitative models

1. Translate three bonds and one lone pair without adding a fifth direction.
2. Choose a model by the output requested, rather than the chapter title.
3. Identify whether the error is in connectivity notation or in its interpretation as physical shape.

Sections

Separate a bonded-atom direction from the number of shared pairs.

The counting centre is nitrogen, not every atom in NF_3 .

Critique the counting operation, rather than merely giving a corrected label.

Steric electron-domain counting and multiple bonds

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

1

Separate a bonded-atom direction from the number of shared pairs.

Medium concept

Section 1 of 10

Next →

Conditions for this construction

- Recognises single and multiple bonds in simple Lewis structures.

This unit builds toward

- Q2 · D2

The move learners miss: Count C-to-O as one local VSEPR domain.

[Try it in Core2](#)

1 Make separate neighbour and shared-pair inventories for HCHO.

Why valid: Two H atoms and one O are attached to C; a C=O double bond contains two shared pairs.

Result: Three neighbours; two pairs in C=O.

2 The step the hard question turns on Count C-to-O as one local VSEPR domain.

Why valid: A multiple bond connects one pair of nuclei and supplies one bonded-neighbour direction in local domain counting. Sigma and pi density have different spatial symmetry.

Result: One C=O domain; this does not place pi density on the internuclear axis.

3 Add the two C-H directions.

Why valid: Each of the two distinct C-H connections adds one domain and C has no lone pair.

Result: $1+1+1 =$ three C-centred domains.

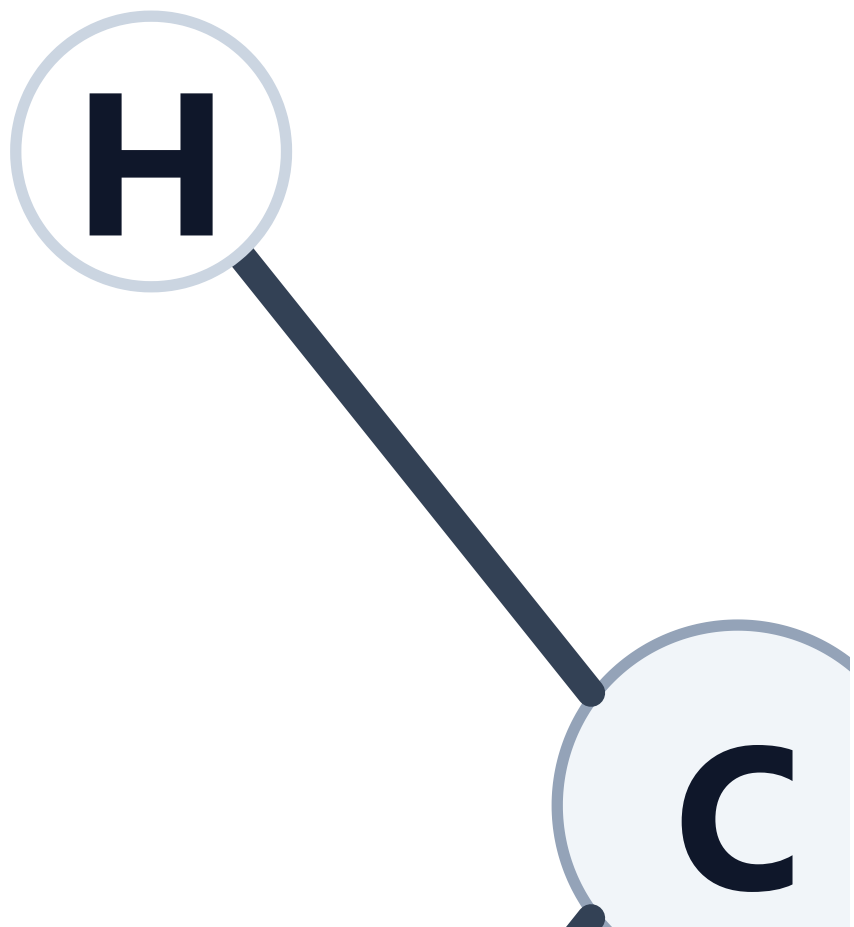
4 Recount carbon directions in HCN.

Why valid: H-C and C-N connect carbon to two neighbours although C-N has three shared pairs.

Result: Two domains in HCN; shared-pair count and neighbour count are different.

Stage 1: Lewi

Four shared electi





Student trap: count
as 4 separate

Identify shared pairs and non-bonding pairs.

Stage 1 of 3

Group bonding and non-bonding electron pairs into spatial steric domains around central atoms.

The key step

HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.

Formula reference • meaning and conditions

Equations and validity

Equation	What it tells you	When you can
Steric Number = (number of bonded atoms) + (number of lone pairs on central atom)	Each bonded atom constitutes exactly one bonding domain regardless of bond order (single, double, triple); only lone pairs localized on the central atom contribute to its steric environment.	- Count each attached atom as one domain regardless of single/double bond order - Include only unshared electron pairs on the central atom, excluding those on ligand lone pairs

Worked explanation

At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.

Follow the operation, its justification and its result.

- The sigma component and pi component of a multiple bond connect the same atom pair.
- HCN: two carbon directions, three C-N pairs. HCHO: three directions, two C-

O pairs.

3. HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.

Result

HCN: two carbon directions, three C-N pairs. HCHO: three directions, two C-O pairs.

Check

Changed check: a triple bond to N in HCN still supplies one C-to-N direction; carbon also has the H direction.

Mistake clinic • diagnose and repair

A tempting wrong path

- Each line of a double bond is a separate local direction.
- Terminal-atom lone pairs add domains at the central atom.
- Bond multiplicity equals neighbour-direction multiplicity.

Diagnose

- At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.
- Count N-centred domains in NH₃ and NF₃. Then state where the F lone pairs are located.
- A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Repair

- HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.
- NF₃ has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.

- Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: a triple bond to N in HCN still supplies one C-to-N direction; carbon also has the H direction.

2 • APPLY

Explain why replacing a double bond with a triple bond to the same neighbour need not add a direction.

3 • CONNECT

The sigma component and pi component of a multiple bond connect the same atom pair.

2

The counting centre is nitrogen, not every atom in NF₃.

Medium concept

Section 2 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q8 • D2

The move learners miss: Assign each lone pair to its own atom.

[Try it in Core2](#)

1 Choose nitrogen as the counting centre.

Why valid: The requested domains are centred on N, not on the entire NF₃ molecule.

Result: Three N-F bonding directions are included.

2 The step the hard question turns on Assign each lone pair to its own atom.

Why valid: N has one lone pair; terminal F lone pairs belong to F-centred inventories.

Result: One N lone pair included; terminal F lone pairs excluded from the N count.

3 Add the local entries.

Why valid: The N inventory has three bonded directions plus one central lone pair.

Result: $3 + 1 =$ four nitrogen-centred domains.

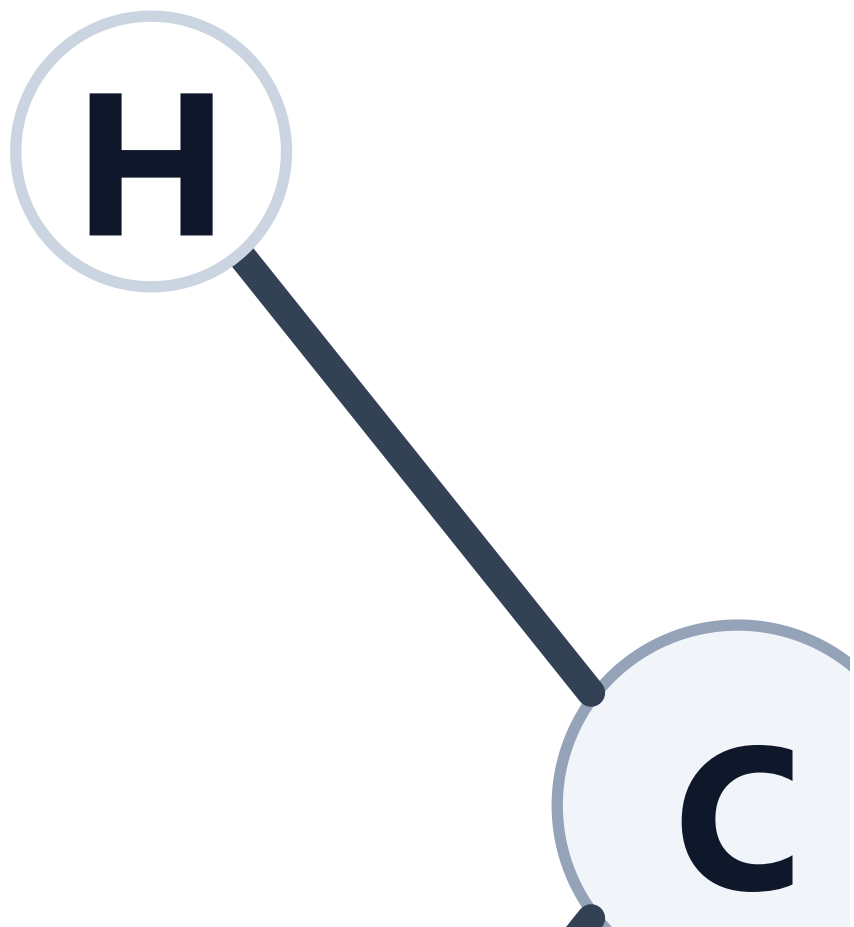
4 Substitute H for F in the local comparison.

Why valid: NH₃ has the same three-bond/one-central-pair N inventory even though terminal atoms differ.

Result: NH₃ also has four N domains; terminal pairs were not required for that total.

Stage 1: Lewi

Four shared electi





Student trap: count
as 4 separate

Identify shared pairs and non-bonding pairs.

Stage 1 of 3

Group bonding and non-bonding electron pairs into spatial steric domains around central atoms.

Worked explanation

Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.

Follow the operation, its justification and its result.

1. Local domains consist of bonds and lone pairs at the selected atom.
2. Both N centres have four domains; F pairs are located at F.
3. NF_3 has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.

Result

Both N centres have four domains; F pairs are located at F.

Check

Changed check: substituting H for F leaves the local three-bond-plus-lone-pair N inventory unchanged in the introductory model.

Mistake clinic • diagnose and repair

A tempting wrong path

- Each line of a double bond is a separate local direction.
- Terminal-atom lone pairs add domains at the central atom.
- Bond multiplicity equals neighbour-direction multiplicity.

Diagnose

- At carbon in HCN , count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO .
- Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.
- A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO .

Repair

- HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.
- NF₃ has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.
- Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: substituting H for F leaves the local three-bond-plus-lone-pair N inventory unchanged in the introductory model.

2 • APPLY

Count domains at oxygen in H₂O while excluding electrons assigned to H.

3 • CONNECT

Local domains consist of bonds and lone pairs at the selected atom.

3

Critique the counting operation, rather than merely giving a corrected label.

Medium concept

Section 3 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q9** · D2

The move learners miss: Replace bond-line counting with local neighbour counting. [Try it in Core2](#)

1 Identify the faulty C=O line-counting entry.

Why valid: Two drawn bond lines denote two bond components to one O neighbour.

Result: The student counted one neighbour twice.

2 The step the hard question turns on Replace bond-line counting with local neighbour counting.

Why valid: In VSEPR a multiple bond supplies one bonded-neighbour domain; pi density is lateral rather than an extra atom direction.

Result: C=O contributes one domain, not two.

3 Compute the corrected HCHO total.

Why valid: There are two distinct H neighbours, one O neighbour and no C lone pair.

Result: $2+1+0 =$ three C-centred domains.

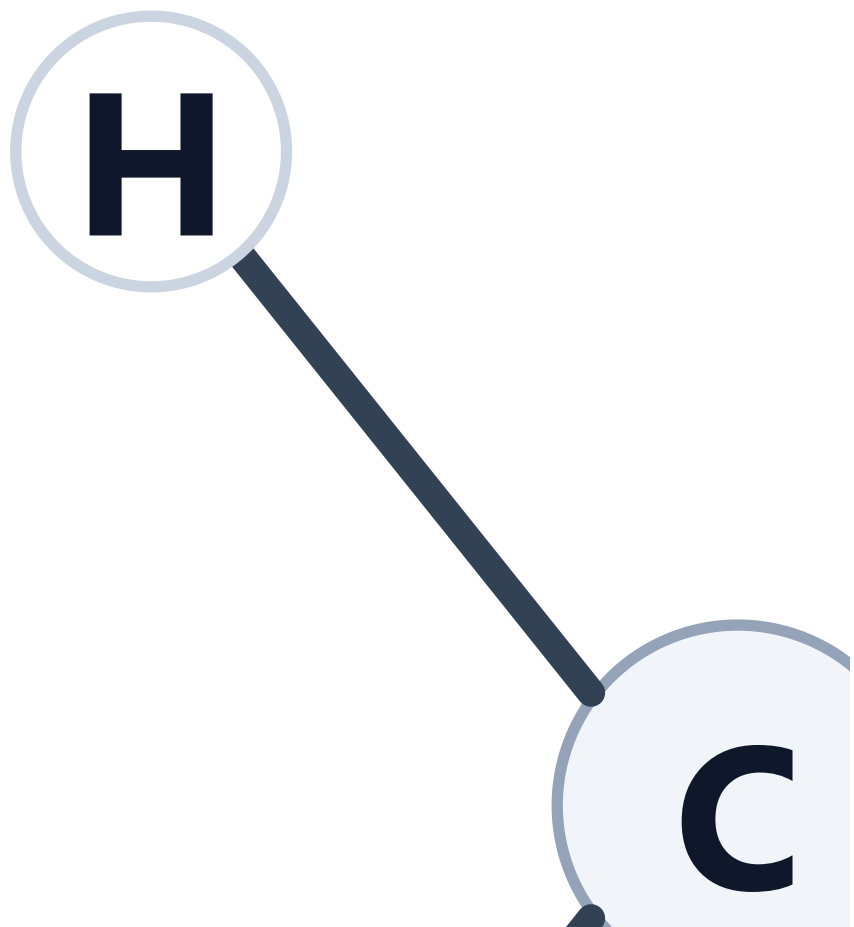
4 Apply the operation to CO₂.

Why valid: Two C=O bonds connect C to two O neighbours; four drawn bond lines do not imply four local domains.

Result: CO₂ carbon has two domains in the introductory model.

Stage 1: Lewi

Four shared electi





Student trap: count
as 4 separate

Identify shared pairs and non-bonding pairs.

Stage 1 of 3

Group bonding and non-bonding electron pairs into spatial steric domains around central atoms.

Worked explanation

A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Follow the operation, its justification and its result.

1. Bond multiplicity and neighbour directions are different ledgers.
2. HCN two; HCHO three. Repeating line-counting in both cases indicates the rule is held rather than a single arithmetic slip.
3. Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

Result

HCN two; HCHO three. Repeating line-counting in both cases indicates the rule is held rather than a single arithmetic slip.

Check

Changed check: HCN carbon has two neighbour directions even though C-N is triple.

Mistake clinic • diagnose and repair

A tempting wrong path

- Each line of a double bond is a separate local direction.
- Terminal-atom lone pairs add domains at the central atom.
- Bond multiplicity equals neighbour-direction multiplicity.

Diagnose

- At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.
- Count N-centred domains in NH₃ and NF₃. Then state where the F lone pairs are located.

- A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Repair

- HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.
- NF₃ has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.
- Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: HCN carbon has two neighbour directions even though C-N is triple.

2 • APPLY

Audit a proposed domain count for CO₂.

3 • CONNECT

Bond multiplicity and neighbour directions are different ledgers.

Question-specific diagnosis and repair

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Repair for Q2

Test the rule on a changed case

At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.

Your prediction and reason

Repair for Q8

Test the rule on a changed case

Count N-centred domains in NH_3 and NF_3 . Then state where the F lone pairs are located.

Your prediction and reason

Repair for Q9

Test the rule on a changed case

A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Your prediction and reason

Now you do one

At carbon in CO_2 , count local domains and then separately count the shared pairs in both $\text{C}=\text{O}$ bonds. Explain why the totals differ.

Model answer

Two carbon-neighbour domains; four shared pairs. Each double bond connects one O neighbour.

1. Two carbon-neighbour domains; four shared pairs. Each double bond connects one O neighbour.

Practice this concept in Core2

Q2

Q8

Q9

Sections

Distinguish the hybrid-set name from the number of orbitals in that set.

The task asks for a local count before the introductory geometry label.

Use the stated internuclear axis to classify the overlap.

Map each set name to an orbital inventory, not its superscript alone.

Hybridisation, orbital conservation, and overlap symmetry

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

1

Distinguish the hybrid-set name from the number of orbitals in that set.

Medium concept

Section 4 of 10

← Previous

Next →

Conditions for this construction

- Understands basic s and p atomic orbital shapes.

Prerequisite reference

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- [Inventory central-atom steric electron domains by grouping multiple bonds into single domains and including only central lone pairs](#)

Why this section matters for the question set

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This unit builds toward

- **Q1 · D1**

The move learners miss: Select one valence s and three valence p inputs.

[Try it in Core2](#)

1 Read the requested four equivalent domain directions.

Why valid: The question supplies a tetrahedral four-direction introductory model.

Result: Four model directions must be represented.

2 The step the hard question turns on Select one valence s and three valence p inputs.

Why valid: For a second-period centre, these four independent basis functions can be recombined without changing the basis dimension.

Result: $2s + 2p_x + 2p_y + 2p_z$: four inputs.

3 Name the recombined set and count its members.

Why valid: The name sp^3 records one s and three p inputs; the total number is $1+3$.

Result: sp^3 ; four hybrid orbitals.

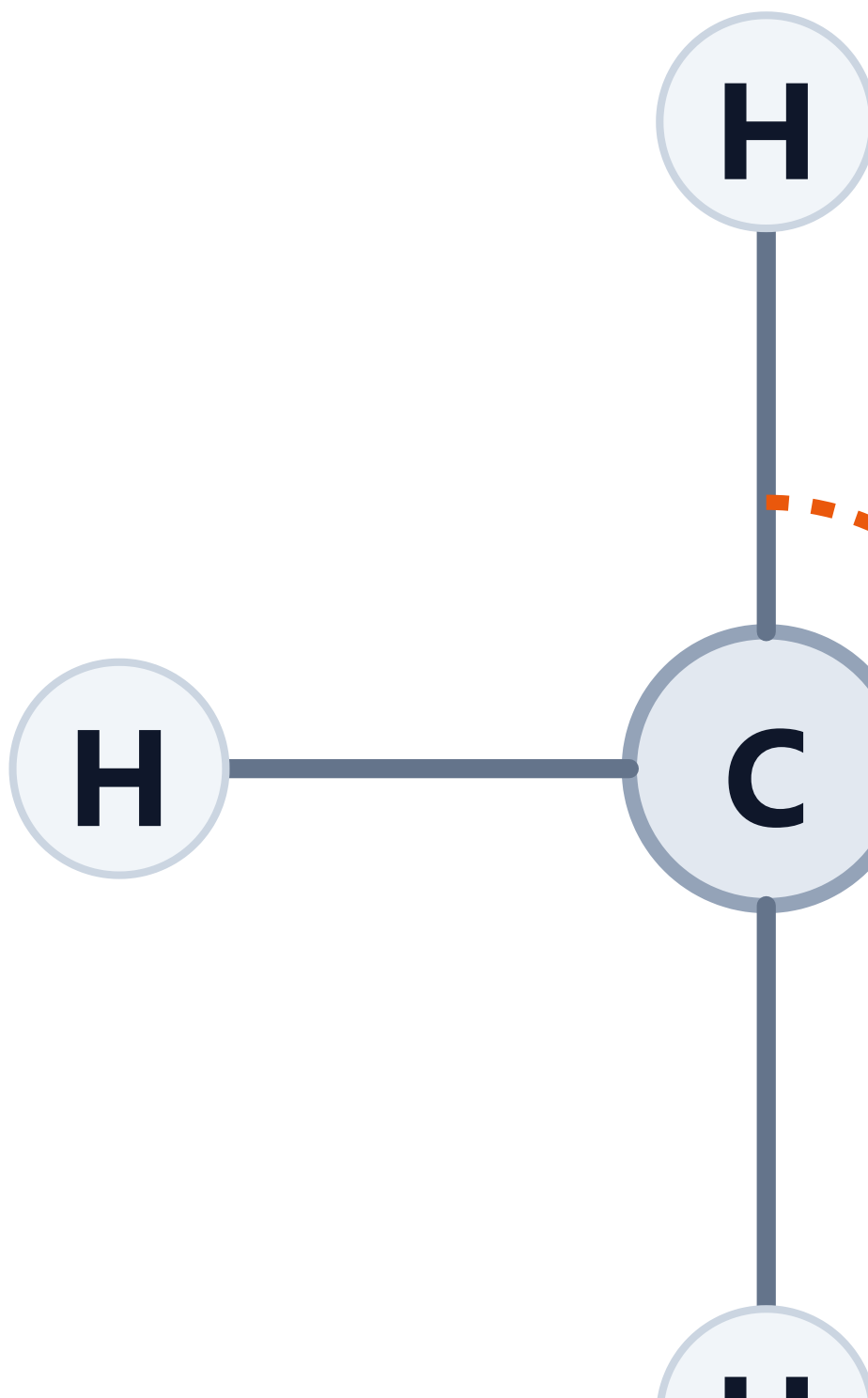
4 Compare the count with an sp allocation.

Why valid: One s plus one p has two outputs, so the superscript alone cannot be the set size.

Result: sp has two outputs; the four-member sp^3 inventory is consistent.

Stage 1: 2D Lewis C

Topological connectivity (F





Misleading: 2D drawing forc

Recognise that flat drawings force artificial 90-degree angles.

Stage 1 of 3

Translate flat 2D Lewis drawings into 3D non-coplanar tetrahedral geometry using wedge-dash notation.

The key step

BF₃: three domains and three sp² hybrids. CH₄: four domains and four sp³ hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

Formula reference · meaning and conditions

Equations and validity

Equation	What it tells you	When you can
n atomic orbitals mixed $\rightarrow$ n hybrid orbitals produced (sp $\rightarrow$ 2, sp ² $\rightarrow$ 3, sp ³ $\rightarrow$ 4)	Linear combination of valence atomic orbitals conserves total orbital count; the number of directional hybrid orbitals formed equals the steric number of electron domains.	Hybridisation occurs because orbitals combine to form lower energy hybrid orbitals.

Equation	What it tells you	When you can
		the central atom. Number of hybrid orbitals formed equals the number of atomic orbitals hybridized.
σ bond: coaxial end-on overlap with cylindrical symmetry; π bond: lateral side-on overlap with nodal plane on internuclear axis	Covalent bonds are classified by orbital overlap geometry: sigma bonds have electron density concentrated directly along the internuclear axis; pi bonds have electron density above and below that axis.	- Internuclear axis is conventionally taken as the z-axis. - Atomic orbitals must have matching symmetry for favorable overlap.

Worked explanation

An sp^2 set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.

Follow the operation, its justification and its result.

1. A change of orbital basis preserves the number of independent orbitals.
2. Three for sp^2 , two for sp . Repeating the superscript-as-total rule in both cases suggests a held counting rule; correct changed cases after an isolated slip suggest an execution error.

3. For second-period atoms, one 2s plus three 2p orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.

Result

Three for sp^2 , two for sp . Repeating the superscript-as-total rule in both cases suggests a held counting rule; correct changed cases after an isolated slip suggest an execution error.

Check

Changed check: an sp set has two hybrids because one s plus one p supplies two basis functions; use this conservation rule on the requested set.

Mistake clinic • diagnose and repair

A tempting wrong path

- The superscript is the total number of hybrid orbitals.
- The page layout is the molecular geometry.
- Turning a drawing on the page changes sigma into pi.
- sp^2 has two hybrids and sp^3 has three.

Diagnose

- An sp^2 set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.
- Count domains at C in CH_3Cl and C in CH_4 . Does redrawing either as a cross change its count?
- Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.
- Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Repair

- For second-period atoms, one 2s plus three 2p orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.
- CH_4 has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp^3 description. A Lewis layout encodes connectivity.

- Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.
- BF₃: three domains and three sp² hybrids. CH₄: four domains and four sp³ hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: an sp set has two hybrids because one s plus one p supplies two basis functions; use this conservation rule on the requested set.

2 • APPLY

Give the input/output inventory for sp without copying the original set.

3 • CONNECT

A change of orbital basis preserves the number of independent orbitals.

2

The task asks for a local count before the introductory geometry label.

Medium concept

Section 5 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q3 · D1

The move learners miss: Apply the explicitly supplied four-domain rule.

[Try it in Core2](#)

1 Inventory CH₄ at carbon.

Why valid: The supplied Lewis connectivity has four C-H bonds and no C lone pair.

Result: Four bonding domains; zero central lone pairs.

2 The step the hard question turns on Apply the explicitly supplied four-domain rule.

Why valid: The task asks for the known introductory correspondence, rather than an experimental determination of orbital coefficients.

Result: Tetrahedral domain geometry; introductory sp^3 description.

3 Check the orbital basis ledger.

Why valid: One 2s plus three 2p inputs gives four hybrids, matching four model sigma directions.

Result: Four inputs and four outputs; no residual p in this allocation.

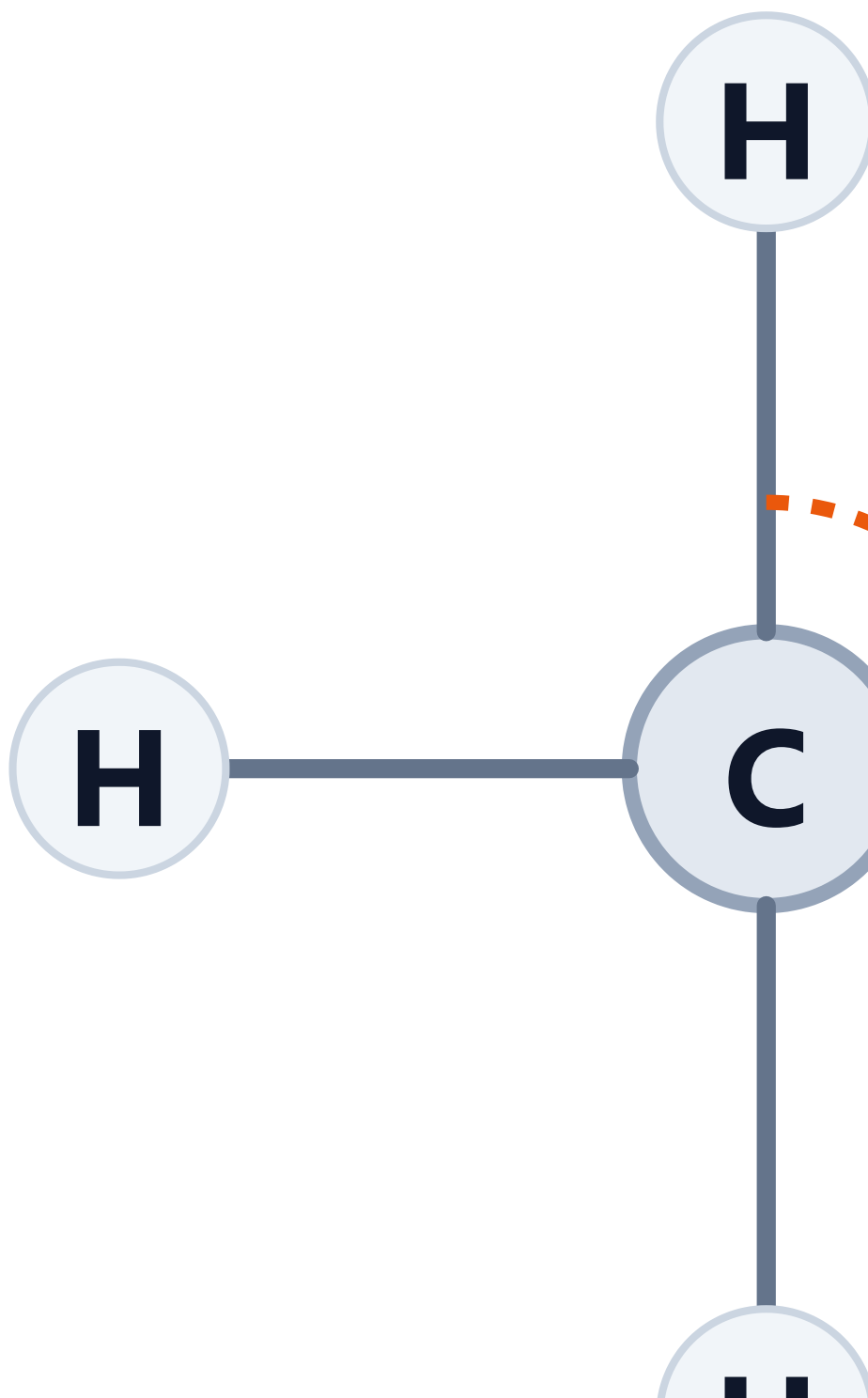
4 Change one substituent without adding a neighbour.

Why valid: CH_3Cl still has four carbon-neighbour connections; different bond properties do not constitute extra domains.

Result: CH_3Cl also has four C-centred domains; no exact bond-property claim follows.

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Recognise that flat drawings force artificial 90-degree angles.

Stage 1 of 3

Translate flat 2D Lewis drawings into 3D non-coplanar tetrahedral geometry using wedge-dash notation.

Worked explanation

Count domains at C in CH_3Cl and C in CH_4 . Does redrawing either as a cross change its count?

Follow the operation, its justification and its result.

1. Four bonding directions and no central lone pair use the demonstrated four-domain rule.
2. Both have four; changing drawing placement changes neither connectivity nor domain count.
3. CH_4 has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp^3 description. A Lewis layout encodes connectivity.

Result

Both have four; changing drawing placement changes neither connectivity nor domain count.

Check

Changed check: CH₃Cl still has four bonded neighbours around carbon even if the bonds have different properties.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- The superscript is the total number of hybrid orbitals.
- The page layout is the molecular geometry.
- Turning a drawing on the page changes sigma into pi.
- sp² has two hybrids and sp³ has three.

Diagnose

- An sp² set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.
- Count domains at C in CH₃Cl and C in CH₄. Does redrawing either as a cross change its count?
- Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.
- Compare proposed totals sp=1, sp²=2, sp³=3 against basis-function conservation. Give the corrected sequence.

Repair

- For second-period atoms, one 2s plus three 2p orbitals gives four sp³ hybrids. The superscript counts p inputs, not total outputs.
- CH₄ has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp³ description. A Lewis layout encodes connectivity.
- Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.
- BF₃: three domains and three sp² hybrids. CH₄: four domains and four sp³ hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: CH_3Cl still has four bonded neighbours around carbon even if the bonds have different properties.

2 • APPLY

Classify carbon in CH_2Cl_2 by domain inventory, not bond identity.

3 • CONNECT

Four bonding directions and no central lone pair use the demonstrated four-domain rule.

3

Use the stated internuclear axis to classify the overlap.

Medium concept

Section 6 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q6** · D2

The move learners miss: Classify A from its stated end-on overlap.

[Try it in Core2](#)

1 Mark the line joining the nuclei in each sketch.

Why valid: Sigma/pi labels refer to overlap relative to this internuclear axis, not to page orientation.

Result: A common geometric reference is declared.

2 The step the hard question turns on Classify A from its stated end-on overlap.

Why valid: The supplied overlap is axial; in the introductory sketch the sigma component is symmetric about the bond axis.

Result: A is sigma overlap.

3 Classify B from side-on p overlap.

Why valid: The supplied parallel lateral lobes have a nodal plane containing

the axis.

Result: B is pi overlap.

4 Rotate the whole picture, including its axis.

Why valid: A rigid rotation preserves all relative orientation and symmetry relations.

Result: Neither classification changes under whole-picture rotation.

Stage 1: Internuclear Axis Reference

The internuclear axis (z-axis) connects the nuclei of

Establish the reference line connecting the nuclei.

Stage 1 of 3

Classify orbital overlaps as coaxial end-on sigma bonds versus lateral side-on pi bonds with nodal planes.

Worked explanation

Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi?

Compare that with rotating one p orbital relative to its neighbour.

Follow the operation, its justification and its result.

1. End-on and side-on are orientation/symmetry descriptions, not molecule-name clues.
2. Whole-picture rotation does not change sigma/pi classification; relative orbital rotation can change overlap. Conflating page orientation with relative geometry suggests the misconception.
3. Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.

Result

Whole-picture rotation does not change sigma/pi classification; relative orbital rotation can change overlap. Conflating page orientation with relative geometry suggests the misconception.

Check

Changed check: rotating the entire sketch on the page leaves its classification unchanged relative to the rotated axis.

Mistake clinic • diagnose and repair

A tempting wrong path

- The superscript is the total number of hybrid orbitals.
- The page layout is the molecular geometry.
- Turning a drawing on the page changes sigma into pi.
- sp^2 has two hybrids and sp^3 has three.

Diagnose

- An sp^2 set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.
- Count domains at C in CH_3Cl and C in CH_4 . Does redrawing either as a cross change its count?
- Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.
- Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Repair

- For second-period atoms, one 2s plus three 2p orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.
- CH_4 has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp^3 description. A Lewis layout encodes connectivity.
- Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.
- BF_3 : three domains and three sp^2 hybrids. CH_4 : four domains and four sp^3 hybrids. For these centres use valence 2s/2p language rather than

ambiguous 1s + 2p shorthand.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: rotating the entire sketch on the page leaves its classification unchanged relative to the rotated axis.

2 • APPLY

Label a sigma component within a triple bond using its geometry.

3 • CONNECT

End-on and side-on are orientation/symmetry descriptions, not molecule-name clues.

4

Map each set name to an orbital inventory, not its superscript alone.

Medium concept

Section 7 of 10

← Previous

Next →

Why this section matters for the question set

This unit builds toward

- Q10 • D2

The move learners miss: Match each inventory to a conserved valence-basis allocation. [Try it in Core2](#)

1 Read the two supplied domain inventories.

Why valid: The task supplies three B bonding domains and four C bonding domains, with no central lone pairs.

Result: BF₃: three; CH₄: four.

2 The step the hard question turns on Match each inventory to a conserved valence-basis allocation.

Why valid: One s plus two p yields three hybrids; one s plus three p yields

four.

Result: BF₃: sp², three hybrids; CH₄: sp³, four hybrids.

3 Match the ideal domain arrangement within this introductory model.

Why valid: Three equivalent model directions are planar; four are tetrahedral. These labels are model descriptions.

Result: BF₃: trigonal planar; CH₄: tetrahedral.

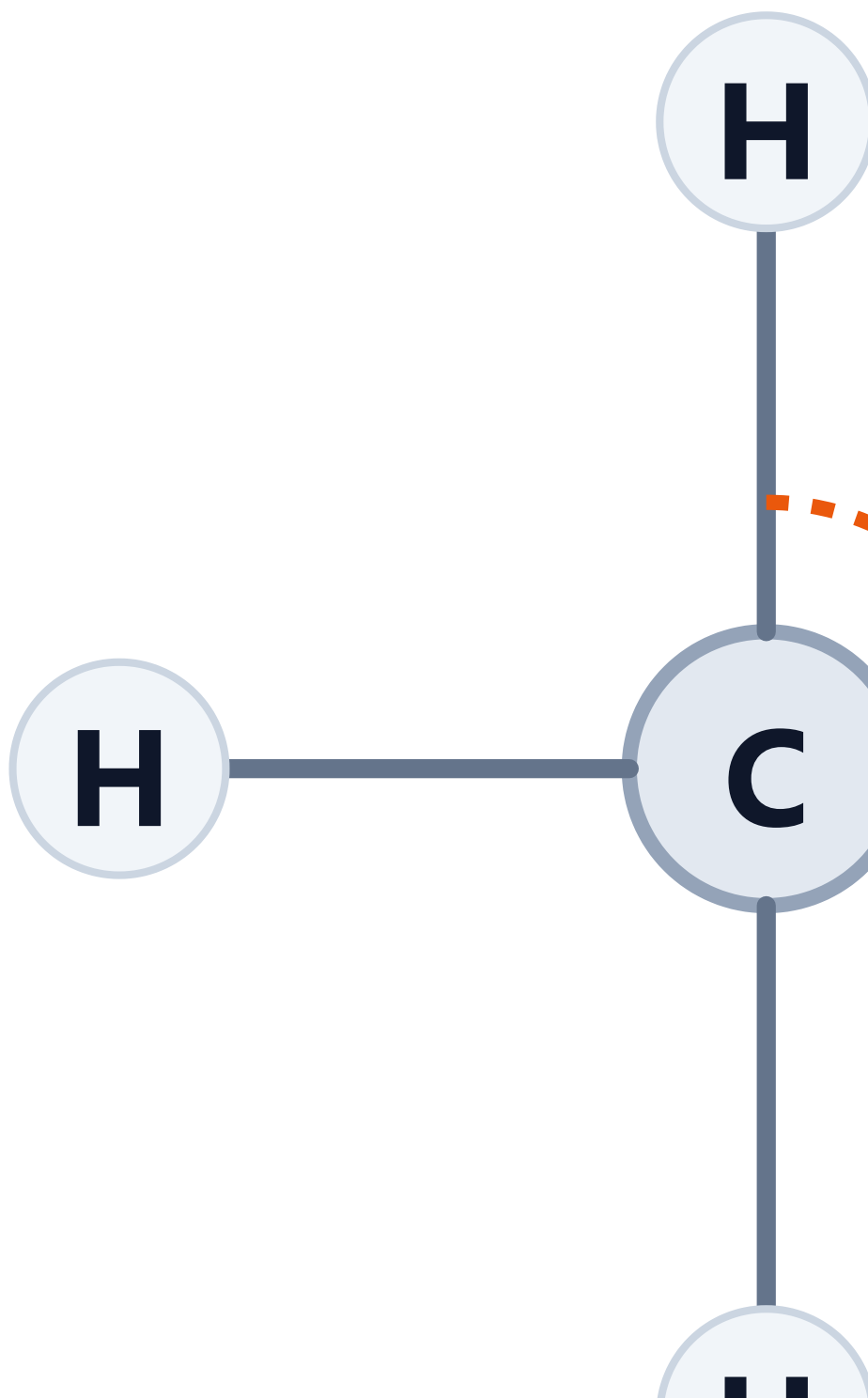
4 Reconstruct an sp inventory as a third case.

Why valid: One s plus one p provides two inputs and therefore two hybrid outputs.

Result: The set-size sequence is sp=2, sp²=3, sp³=4, rather than the superscripts 1,2,3.

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Recognise that flat drawings force artificial 90-degree angles.

Stage 1 of 3

Translate flat 2D Lewis drawings into 3D non-coplanar tetrahedral geometry using wedge-dash notation.

Worked explanation

Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Follow the operation, its justification and its result.

1. One s plus n p inputs gives $n+1$ hybrid outputs.
2. The corrected sequence is 2,3,4. Repeating the shifted sequence suggests the superscript-as-total rule.
3. BF_3 : three domains and three sp^2 hybrids. CH_4 : four domains and four sp^3 hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

Result

The corrected sequence is 2,3,4. Repeating the shifted sequence suggests the superscript-as-total rule.

Check

Changed check: sp uses two inputs and two hybrid outputs, providing a third case for the conservation rule.

Mistake clinic • diagnose and repair

A tempting wrong path

- The superscript is the total number of hybrid orbitals.
- The page layout is the molecular geometry.
- Turning a drawing on the page changes sigma into pi.
- sp^2 has two hybrids and sp^3 has three.

Diagnose

- An sp^2 set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.
- Count domains at C in CH_3Cl and C in CH_4 . Does redrawing either as a cross change its count?
- Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.
- Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Repair

- For second-period atoms, one 2s plus three 2p orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.
- CH_4 has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp^3 description. A Lewis layout encodes connectivity.
- Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.
- BF_3 : three domains and three sp^2 hybrids. CH_4 : four domains and four sp^3 hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: sp uses two inputs and two hybrid outputs, providing a third case for the conservation rule.

2 • APPLY

Reconstruct all three set sizes without memorising the sequence.

3 • CONNECT

One s plus n p inputs gives n+1 hybrid outputs.

Question-specific diagnosis and repair

Repair for Q1

Test the rule on a changed case

An sp² set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.

Your prediction and reason

Repair for Q3

Test the rule on a changed case

Count domains at C in CH₃Cl and C in CH₄. Does redrawing either as a cross change its count?

Your prediction and reason

Repair for Q6

Test the rule on a changed case

Rotate the whole end-on sketch by 90 degrees on paper. Does it become π ?
Compare that with rotating one p orbital relative to its neighbour.

Your prediction and reason

Repair for Q10

Test the rule on a changed case

Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Your prediction and reason

Now you do one

For a supplied two-domain second-period centre with no central lone pair, reconstruct the introductory hybrid set size and unused p inventory.

Model answer

An sp allocation has two hybrids and two residual p functions; four valence-basis functions are accounted for.

1. An sp allocation has two hybrids and two residual p functions; four valence-basis functions are accounted for.

Practice this concept in Core2

Q1

Q3

Q6

Q10

Sections

Translate three bonds and one lone pair without adding a fifth direction.

Choose a model by the output requested, rather than the chapter title.

Identify whether the error is in connectivity notation or in its interpretation as physical shape.

3D Stereochemical representation and qualitative models

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

1

Translate three bonds and one lone pair without adding a fifth direction.

Medium concept

Section 8 of 10

← Previous

Next →

Conditions for this construction

- Understands that molecules occupy 3D space.

Prerequisite reference

- [Inventory central-atom steric electron domains by grouping multiple bonds into single domains and including only central lone pairs](#)
- [Map steric electron domain counts to introductory hybrid-orbital sets \(sp, sp², sp³\) and classify coaxial vs lateral orbital overlaps](#)

This unit builds toward

- **Q4 · D2**

The move learners miss: Project the four domain directions with a declared depth convention. [Try it in Core2](#)

1 Count exactly three N-H connections and one N lone pair.

Why valid: The question supplies the inventory at nitrogen.

Result: Four domains: three bonds and one lone pair.

2 The step the hard question turns on Project the four domain directions with a declared depth convention.

Why valid: Two ordinary lines may denote in-plane directions; a solid wedge denotes toward and a hashed wedge away. The lone-pair slot must replace one bond slot.

Result: N at centre; exactly three labelled H endpoints and one LP direction.

3 Distinguish domain arrangement from nuclear shape.

Why valid: Domain geometry counts all four slots; molecular shape counts the three bonded H nuclei.

Result: Tetrahedral domain model; trigonal-pyramidal molecular shape, without an exact-angle inference.

4 Compare with the four-bond NH₄⁺ inventory.

Why valid: Replacing the LP slot with a fourth N-H bond adds a bonded nucleus but no fifth domain.

Result: NH₃: three H + LP; NH₄⁺: four H. The NH₃ drawing must not contain a fourth H.

NH₃: three N-H bond

Use supplied givens only.

Stage 1 of 3

Construct exactly three N-H domains and one distinguishable lone-pair domain.

The key step

NH₃ has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

Formula reference • meaning and conditions

—

Equations and validity

Equation	What it tells you	When you can use it
$\text{Steric Number} = (\text{number of bonded atoms}) + (\text{number of lone pairs on central atom})$	Each bonded atom constitutes exactly one bonding domain regardless of bond order (single, double, triple); only lone pairs localized on the central atom contribute to its steric environment.	- Count each attached atom as one domain regardless of bond order (single/double/triple). - Include only unshared electron pairs on the central atom, excluding terminal ligand lone pairs.
$n \text{ atomic orbitals mixed} \rightarrow n \text{ hybrid orbitals produced}$ (sp → 2, sp² → 3, sp³ → 4)	Linear combination of valence atomic orbitals conserves total orbital count; the number of directional hybrid orbitals formed equals the steric number of electron domains.	- Hybridisation occurs between orbitals of comparable energy in the valence shell of the central atom. - Number of hybrid orbitals formed is strictly equal to the number of atomic orbitals hybridized.

Worked explanation

Draw NH₃ and a tetrahedral four-bond NH₄⁺ comparison. Mark which slot differs and recount the H atoms in each.

Follow the operation, its justification and its result.

1. Wedges and dashes encode depth; electron-domain geometry includes lone pairs, molecular shape uses nuclei.
2. NH_3 has three H plus a lone pair; NH_4^+ has four H and no N lone pair in the local Lewis model.
3. NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

Result

NH_3 has three H plus a lone pair; NH_4^+ has four H and no N lone pair in the local Lewis model.

Check

Count all four slots in the sketch; replacing the lone-pair slot by an N-H bond changes the inventory to an NH_4^+ teaching comparison, not NH_3 .

Mistake clinic • diagnose and repair

A tempting wrong path

- Adding depth marks permits an extra bond or domain.
- A familiar chapter keyword selects the model.
- A Lewis cross proves a square-planar molecule.

Diagnose

- Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.
- Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.
- Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Repair

- NH₃ has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.
- VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.
- For CH₄ a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

1-2-3 QUICK CHECK

1 • CHECK

Count all four slots in the sketch; replacing the lone-pair slot by an N-H bond changes the inventory to an NH₄⁺ teaching comparison, not NH₃.

2 • APPLY

Create an H₂O domain sketch with two bonded slots and two lone-pair slots.

3 • CONNECT

Wedges and dashes encode depth; electron-domain geometry includes lone pairs, molecular shape uses nuclei.

2

Choose a model by the output requested, rather than the chapter title.

Medium concept

Section 9 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q5** · D3 · the hardest question in this set
The move learners miss: Choose VSEPR for that output. [Try it in Core2](#)

1 Specify the required output before choosing a model.

Why valid: The task asks for qualitative domain arrangement and expressly excludes detailed orbital overlap.

Result: Required output: qualitative three-dimensional domain geometry.

2 The step the hard question turns on Choose VSEPR for that output.

Why valid: VSEPR uses the three bonding domains and central lone pair to describe their approximate spatial arrangement.

Result: Four-domain, approximately tetrahedral arrangement; pyramidal bonded-nucleus shape.

3 Declare unresolved quantities.

Why valid: A qualitative domain model does not supply orbital wavefunctions, overlap amplitudes or exact measured angles.

Result: No detailed orbital composition, exact angle or spectrum is determined.

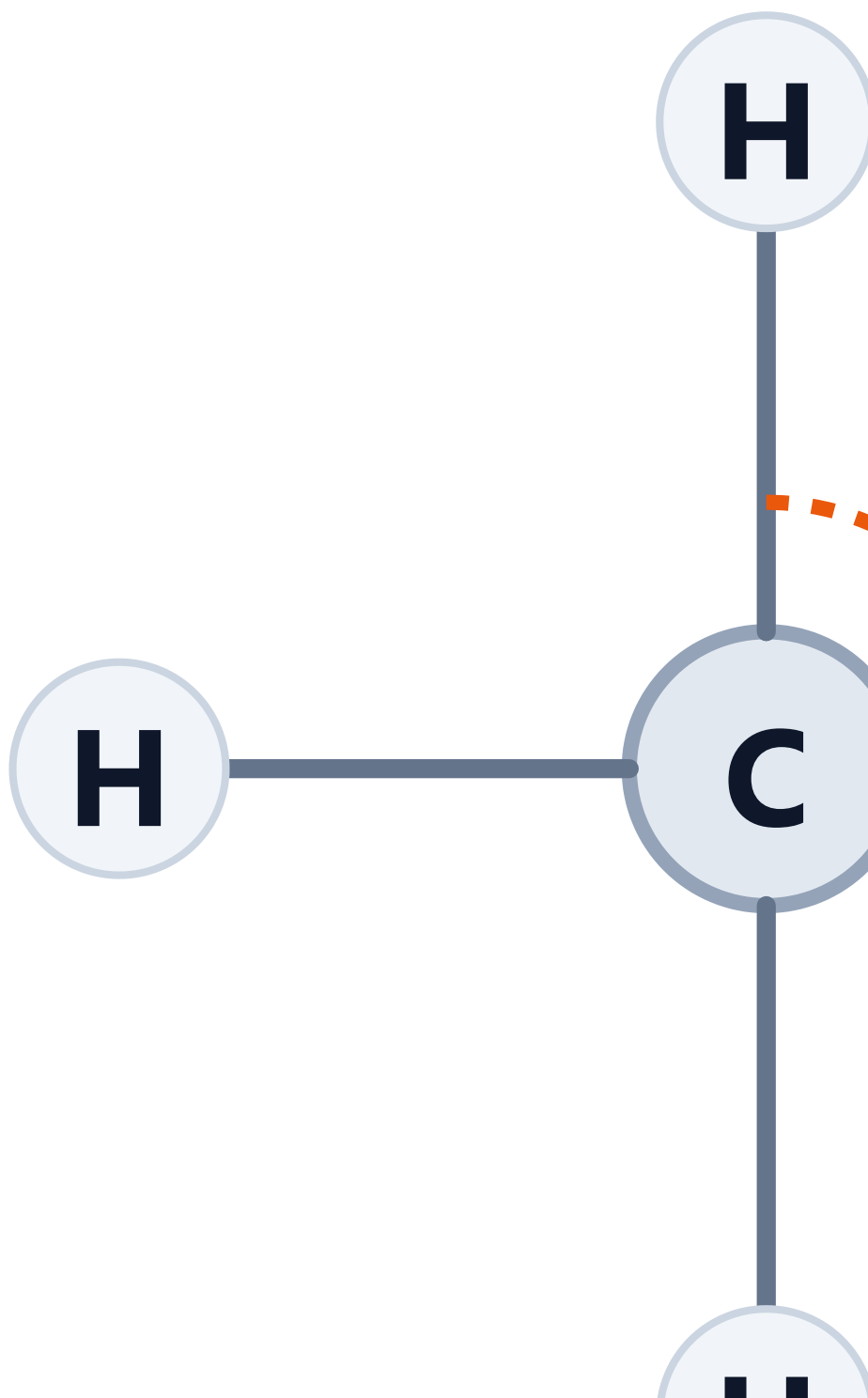
4 Change the required output to axial versus lateral overlap.

Why valid: That question needs orbital orientation/symmetry information beyond a geometry-only model.

Result: An orbital-overlap account is now required; familiarity alone cannot select the model.

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Recognise that flat drawings force artificial 90-degree angles.

Stage 1 of 3

Translate flat 2D Lewis drawings into 3D non-coplanar tetrahedral geometry using wedge-dash notation.

Worked explanation

Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

Follow the operation, its justification and its result.

1. A model has a defined prediction scope and unresolved questions.
2. A needs a geometry model; B needs an orbital account. Choosing the same familiar model without checking output in both cases suggests scope confusion.
3. VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.

Result

A needs a geometry model; B needs an orbital account. Choosing the same familiar model without checking output in both cases suggests scope confusion.

Check

Changed task: detailed axial versus side-on overlap now requires an overlap description; a geometry-only account is insufficient.

Mistake clinic • diagnose and repair

A tempting wrong path

- Adding depth marks permits an extra bond or domain.
- A familiar chapter keyword selects the model.
- A Lewis cross proves a square-planar molecule.

Diagnose

- Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.
- Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.
- Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Repair

- NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.
- VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.
- For CH_4 a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

1-2-3 QUICK CHECK

1 • CHECK

Changed task: detailed axial versus side-on overlap now requires an overlap description; a geometry-only account is insufficient.

2 • APPLY

Choose the needed model when the new task asks for pi-overlap orientation in ethene.

3 • CONNECT

A model has a defined prediction scope and unresolved questions.

3

Identify whether the error is in connectivity notation or in its interpretation as physical shape.

Medium concept

Section 10 of 10

← Previous

Why this section matters for the question set

—

This unit builds toward

- **Q7 • D2**

The move learners miss: Declare and use a wedge/dash convention.

[Try it in Core2](#)

1 Retain the four C-H connections from Lewis notation.

Why valid: A Lewis cross records connectivity; its page angles do not measure molecular angles.

Result: C at centre with four H endpoints; no bond is added or removed.

2 The step the hard question turns on **Declare and use a wedge/dash convention.**

Why valid: A solid wedge points toward the viewer, a hashed wedge away; ordinary lines lie in the chosen page plane.

Result: Two in-plane bonds, one toward bond and one away bond represent a tetrahedral model.

3 Explain the physical and projected angles separately.

Why valid: The ideal tetrahedral model has about 109.5-degree physical angles, while a projection distorts page angles.

Result: The flat cross does not assert square-planar CH₄; depth belongs to the 3D interpretation.

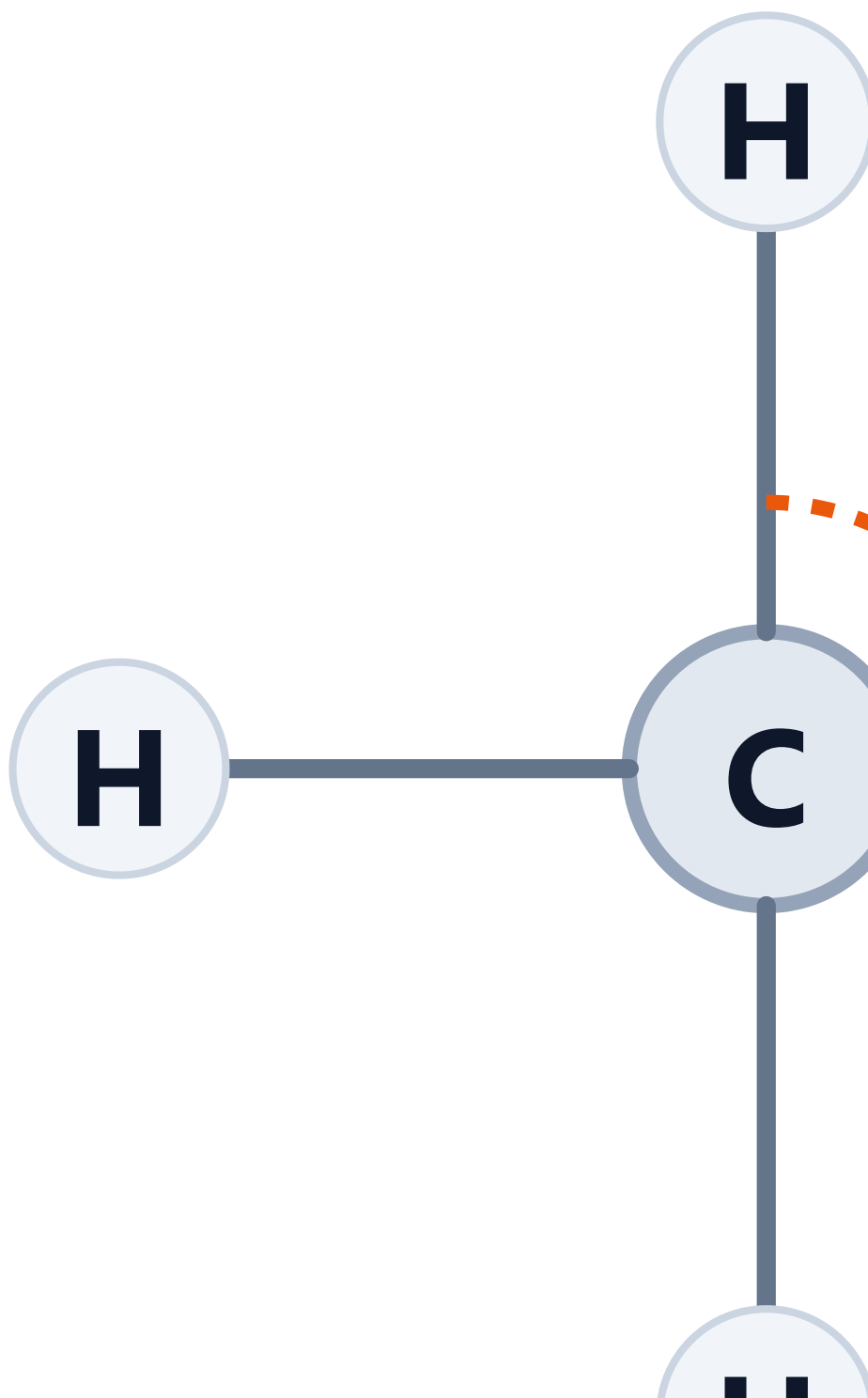
4 Change viewpoint and recount endpoints.

Why valid: A rotation or a different projection preserves connectivity and noncoplanar geometry.

Result: Four C-H bonds remain; the drawing is a projection of the same tetrahedron.

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Recognise that flat drawings force artificial 90-degree angles.

Stage 1 of 3

Translate flat 2D Lewis drawings into 3D non-coplanar tetrahedral geometry using wedge-dash notation.

Worked explanation

Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Follow the operation, its justification and its result.

1. A flat Lewis drawing does not assert that all nuclei are coplanar.
2. No and no.
3. For CH_4 a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

Result

No and no.

Check

Changed check: rotate the paper or choose another viewing direction; the atom connectivity and noncoplanar model must stay unchanged.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Adding depth marks permits an extra bond or domain.
- A familiar chapter keyword selects the model.
- A Lewis cross proves a square-planar molecule.

Diagnose

- Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.
- Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.
- Does a cross-shaped Lewis CH_4 drawing assert square-planar CH_4 ? Does adding wedge/dash notation change the number of bonds?

Repair

- NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.
- VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.
- For CH_4 a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: rotate the paper or choose another viewing direction; the atom connectivity and noncoplanar model must stay unchanged.

2 • APPLY

Explain two different projections of the same tetrahedral model.

3 • CONNECT

A flat Lewis drawing does not assert that all nuclei are coplanar.

Question-specific diagnosis and repair

Repair for Q4

Test the rule on a changed case

Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.

Your prediction and reason

Repair for Q5

Test the rule on a changed case

Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

Your prediction and reason

Repair for Q7

Test the rule on a changed case

Does a cross-shaped Lewis CH₄ drawing assert square-planar CH₄? Does adding wedge/dash notation change the number of bonds?

Your prediction and reason

Now you do one

Draw a depth-labelled domain model for H₂O: two O-H bonds and two central lone pairs. Separate domain arrangement from bonded-nucleus shape.

Model answer

Four O-centred domains: two bonds and two lone pairs. Introductory tetrahedral domain arrangement; bent molecular shape. Exactly two H endpoints.

1. Four O-centred domains: two bonds and two lone pairs. Introductory tetrahedral domain arrangement; bent molecular shape. Exactly two H endpoints.

Practice this concept in Core2

Q4

Q5

Q7